

Chapter 13: Azure Functions

Learning Objectives

By the end of this chapter, you will:

- Understand the fundamentals of serverless computing and Azure Functions.
- Learn how Azure Functions orchestrate workflows in enterprise AI applications.
- Identify how AI TPMs can leverage Azure Functions to build scalable and cost-efficient AI solutions.

Beginner Explanation

Imagine hiring workers only when work arrives.

- No work? No payment.
- More work? Hire more workers automatically.

This is the idea behind **Serverless Computing**.

Azure Functions is Microsoft's serverless service that executes code when an event occurs.

Examples of Events

Event	Example
HTTP Request	User submits a question
Timer	Run a job every morning at 8 AM
Blob Upload	New document uploaded
Queue Message	Process incoming tasks
Event Grid	React to application events

Simple Example

A user asks:

"Summarize this PDF."

The workflow may look like:

1. User uploads PDF.
2. Azure Function triggers.
3. Function extracts text.
4. Azure OpenAI summarizes content.
5. Result is returned to the user.

The user sees a simple experience, but Azure Functions coordinate everything behind the scenes.

Intermediate Explanation

Azure Functions are commonly used as the orchestration layer for enterprise AI applications.

Why Enterprises Use Azure Functions

Benefit	Description
Scalability	Automatically scales with demand
Cost Efficiency	Pay only when executed
Integration	Works with Azure services
Simplicity	Minimal infrastructure management
Flexibility	Supports multiple languages

Common AI Workflows

Workflow	Description
Document Processing	Extract and summarize content
RAG Pipelines	Orchestrate retrieval workflows
AI Agents	Execute tools and actions
Notifications	Send emails and alerts
Batch Jobs	Run scheduled evaluations

Execution Flow

``text id="zpvsI0" User Request ↓ Azure Function ↓ Azure OpenAI ↓ Azure AI Search ↓ Response

```
### Trigger Types

| Trigger | Enterprise Example |
|-----|-----|
| HTTP | Chat applications |
| Blob | PDF processing |
| Queue | Background jobs |
| Timer | Daily reports |
| Event Grid | Workflow automation |

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## Why This Matters for an AI TPM
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Responsibility	Impact
-----	-----
Delivery	Enables rapid delivery of scalable AI workflows.
Risk	Reduces operational complexity and infrastructure management.
Stakeholder Management	Helps explain automation capabilities to business teams.
Budget	Provides cost-efficient execution for event-driven workloads.

Enterprise Example

Business Problem

A logistics company received more than 50,000 shipping documents each month.

Manual processing resulted in:

- Delays
- Human errors
- Increased costs

Solution

Implement an event-driven architecture using Azure Functions.

Architecture

- Azure Blob Storage
- Azure Functions
- Azure OpenAI
- Azure AI Search
- Power BI

Workflow

1. Document uploaded.
2. Azure Function triggers.
3. Text extraction performed.
4. Azure OpenAI summarizes content.
5. Dashboard updated.

Outcome

Metric	Result
-----	-----
Processing Time	Reduced by 85%
Manual Effort	Reduced by 70%
Monthly Savings	~\$60,000
Accuracy	95%

Lessons Learned

- Event-driven architectures improve scalability.
- Monitoring is essential.
- Queue-based processing improves reliability.
- Cost management should be tracked continuously.

Azure Implementation

Services Used

Service	Purpose
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Azure Functions	Workflow orchestration
Azure OpenAI	Content generation
Azure AI Search	Retrieval services
Azure AI Foundry	Evaluations and management

Additional Services

- Azure Blob Storage
- Azure Queue Storage
- Azure Monitor
- Azure Application Insights
- Azure Key Vault

Implementation Workflow

Step 1: Select Trigger Type

Examples:

Trigger	Use Case
-----	-----
HTTP	Chat applications
Blob	File processing
Queue	Batch processing
Timer	Scheduled jobs

Step 2: Configure Function

Define:

- Runtime
- Memory

- Timeout
- Authentication

Step 3: Integrate Services

Connect:

- Azure OpenAI
- Azure AI Search
- Storage Accounts
- Monitoring

Step 4: Implement Observability

Track:

- Execution count
- Failures
- Latency
- Costs
- User activity

Step 5: Scale

Validate:

- Peak loads
- Cold starts
- Retry mechanisms
- Error handling

Mermaid Diagram

```
```mermaid id="b6axv5"  
graph TD
```

```
A[User Request]
```

```
A --> B[Azure Function]
```

```
B --> C[Azure OpenAI]
```

```
C --> D[Azure AI Search]
```

D --> E[Blob Storage]

E --> F[Response]

F --> G[User]

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## Event-Driven Architecture

```
```mermaid id="b3t3o8" graph LR
```

A[Blob Upload]

A --> B[Azure Function]

B --> C[Queue]

C --> D[Azure OpenAI]

D --> E[Power BI Dashboard]

E --> F[Operations Team] ```

TPM Checklist

- ☐ Identify workflow requirements.
- ☐ Select trigger types.
- ☐ Estimate monthly execution costs.
- ☐ Configure monitoring.
- ☐ Conduct performance testing.
- ☐ Define retry mechanisms.
- ☐ Implement security controls.
- ☐ Review scalability requirements.
- ☐ Document runbooks.
- ☐ Obtain stakeholder approval.

Templates

Azure Functions Readiness Assessment

Question	Status
Are triggers defined?	Yes/No
Is monitoring configured?	Yes/No

Question	Status
Are costs estimated?	Yes/No
Is security approved?	Yes/No
Are KPIs established?	Yes/No

Monthly Dashboard

Metric	Target
Success Rate	>99%
Latency	<2 Seconds
Cost	Within Budget
Availability	99.9%
Error Rate	<1%

Risk Register

Risk	Impact	Mitigation
Function Failure	High	Retry policies
High Costs	Medium	Budget alerts
Cold Starts	Medium	Premium plans
Security Issues	High	RBAC and Key Vault

Common Mistakes

1. Choosing the wrong trigger type.
2. Ignoring retry mechanisms.
3. Neglecting monitoring.
4. Underestimating execution costs.
5. Failing to secure endpoints.
6. Ignoring cold start impacts.
7. Not testing under production loads.

Key Takeaways

- Azure Functions provide a scalable, serverless platform for orchestrating AI workflows.
- Event-driven architectures improve efficiency and reduce infrastructure management.

- AI TPMs should understand triggers, costs, and scalability considerations.
 - Monitoring and observability are critical for production workloads.
 - Azure Functions integrate seamlessly with Azure OpenAI and Azure AI Search.
 - Serverless architectures enable organizations to build cost-effective AI applications at scale.
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Footer

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